

Homework 2 Part 2

Face Classification & Verification using CNN

11-785: INTRODUCTION TO DEEP LEARNING (FALL 2026)

Schedule

Release

September 18, 2026

- This homework is out of 100 points: Checkpoint submission + On-Time submission + Canvas quiz
- Join the Kaggle competition ([link](#))

Checkpoint Submission

October 2, 2026, 11:59 PM EST

- Complete HW2P2 **Starter** notebook and submit to **Kaggle**. Achieve very low cutoff.
 - Complete the **Homework 2 Quiz** on Canvas, achieve **FULL** score on MCQs.
- ALL** above requirements must be met to achieve full points. Otherwise, your final score will be lowered by **3%**. ([Click here for checkpoint submission instructions](#))

On-Time Submission

October 9, 2026, 11:59 PM EST

- Cutoff scores will be released on Piazza after the checkpoint deadline.
[Click here to see P2 Scoring Details](#)
 - Submit final scores to **Kaggle Competition** ([link](#)) by **October 9 2026, 11:59 PM EST**.
 - Clean up your code notebook, create the final on-time submission zip package, and submit the zip file to Gradescope ([link](#)) by **October 11, 2026, 11:59 PM EST**.
- [Click here for on-time submission instructions](#)

Slack Submission

October 19, 2026, 11:59 PM EST

- Join and submit your final predictions to the **SLACK Kaggle Competition** ([link](#)).
(access opens only after the on-time deadline)
 - Submit to Gradescope for slack submission ([link](#)) with the same submission instructions.
- You are in charge of keeping track of the slack days used, you have a maximum of 10 days.

Start Here

- **Collaboration policy:**

- You are expected to comply with the [University Policy on Academic Integrity and Plagiarism](#).
- You can share ideas but **not** code, you must submit your own code. All submitted code will be compared against all other submissions made this and previous semesters using [MOSS](#).
- You are allowed to help your friends debug, however, you are **not** allowed to type code for them.
- You are **not** allowed to look at your friends' code while typing your solution.
- You are **not** allowed to directly copy and paste solutions from any external sources.
- Meeting regularly with your study group to work together is highly encouraged. We advise a “divide and conquer” approach **within your study group** to explore different strategies before piecing together the most effective one. However, the actual code used to make the submissions **must be entirely your own**.

- **Overview:**

- **HW2P2:** This section of the homework is intended on developing skills in implementing a Convolution Neural Network from scratch for first, an image classification task on human faces and, second, face verification through transfer learning (more details in [Homework objective](#)). We will test the implementation through an open-ended Kaggle competition, a popular service for hosting predictive modeling and data analytics competitions. The competition link can be found below.
- **HW2P2 Quiz:** You need to take an additional quiz with multiple choices, specific for HW2P2. This quiz can be found on Canvas under Quizzes. It is **mandatory** to complete this quiz before the checkpoint deadline for HW2P2.

- **Submission:** There are multiple submissions for HW2P2, please make sure you understand each one.

- **Canvas Quiz:** [Quiz Link](#)
- **Kaggle:** [Competition Link](#)
- **Gradescope:** [Code Submission Link](#)

The follow submissions on apply **ONLY** when you have missed the on-time deadline -

- **Kaggle (Slack):** [Slack Competition Link](#)

- **Important Notes:**

- **[Parameter limit]:** The maximum number of parameters allowed for HW2P2 is **30 million**.
- **[Model constraints]:** You must **implement your own CNN from scratch**, meaning that your model should be built directly using components from `torch.nn` (or similar), without pre-trained weights. Using models directly from PyTorch's `torchvision.models` is strictly **NOT ALLOWED**.
- **[Disallowed architectures]:** No RNNs (GRUs/LSTMs), Transformers, Graph Neural Networks, or any pretrained encoder/decoder models, etc. Essentially anything other than a CNN.

Homework objective

By the end of this homework, you'll have a better understanding of how face recognition systems work, particularly in distinguishing between different faces, with an emphasis on the functionalities of CNNs and transfer learning. We will use CNNs to extract unique identifying features from images of faces (the classification task) and then learn from that to distinguish between different face (the verification task). Through this exercise, you will specifically learn how:

- To implement CNNs for face images for 2 tasks - classification and verification
 - How to handle image data
 - How to use data augmentation techniques for images
 - How to implement your own CNN architecture
 - How to effectively train and evaluate a CNN model
- To derive meaningful representations
 - How to implement CNNs for representation learning
 - How to define what similarity means in the context of images
 - How to measure meaningful or contextual similarity between two derived representations of images of faces, by generating discriminative and generalizable feature representations of the data
 - Learn how classification and verification tasks are connected through transfer learning
- To explore different architectures and hyperparameters for the optimal solution
 - How to identify various design/architecture choices and hyperparameters that affect your solution
 - How to develop strategies to tune hyperparameters through a technically *infinite* search space to find a good or the best solution
- To engineer the solution using open source tools
 - How to use objects from the PyTorch (GPU)/JAX (TPU) framework to build a CNN
 - How to deal with the challenges in data loading, memory usage, arithmetic precision, numerical stability, initialization techniques, and many more! Ideally, you will learn how to maximize the space and time efficiency for your training and inference
 - How to train and test your models on different cloud platforms like Google Colab, Google Cloud Platform (GCP), Amazon Web Services (AWS), Kaggle and Pittsburgh Super Computer (PSC)

Checklist

Here is a checklist page that you can use to keep track of your progress as you go through the write-up and implement the corresponding sections in your starter notebook.

1. Getting started
 - Join the Kaggle competition
 - Download the starter notebook
 - Decide on a JAX or PyTorch implementation
 - Load the libraries, install Kaggle API, and download Kaggle dataset files
2. Complete the dataset classes, initialize the datasets and data loaders
 - Explore and visualize your dataset to understand its distribution
 - Explore and experiment with transformations and normalization
3. Build your CNN architecture using PyTorch
4. Run training and evaluation
 - Implement the train function for 1 epoch
 - Implement the evaluate function for 1 epoch
 - Implement the save and load functions for backing up your model checkpoints, in case training crashes abruptly
 - Run your training loop for the number of epochs defined in your config file. Make sure you are saving your best performing checkpoints either on WandB and/or locally
5. Test your best performing model during training on the classification task and store this result for your final submission to Kaggle
6. Test how the same model performs on the verification task
7. If there is room for improvement on the verification task, retrain your model on additional data and/or use different loss functions. Test your new model on both classification and verification tasks again.
8. Use your best predictions from the classification and verification tasks to make your final Kaggle submission

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1 Introduction

1.1 Overview

In this homework, we will build and train Convolutional Neural Networks (CNNs) for face recognition tasks. The most successful face recognition systems, as of early 2019 [1, 2], can achieve more than 99% accuracy for faces of any identity. This remarkable accuracy implies that face recognition systems do not necessarily need to be trained on face images of all possible identities in the world but can still robustly conduct **open-set** inference, i.e., recognizing identities not present in the training set. Beyond face recognition, such systems can be extended to many other applications, such as retail product recognition, surveillance person identification, and vehicle identification, as long as they are trained on the specific data relevant to these tasks. In this homework, we will learn how to build our own face recognition systems from two perspectives:

- How to build effective CNN architectures for image classification.
- How to build CNNs for face verification using **transfer learning** on models trained for image classification.

1.2 Architectures of CNNs

Convolutional Neural Networks (CNNs) have significantly evolved since their inception, becoming more sophisticated and efficient in handling complex tasks. Pioneering architectures like LeNet [3] and AlexNet [4] paved the way, focusing on simple layers and large fully connected networks. Subsequent architectures, such as VGGNet [5], introduced deeper networks with uniform layer structures, while ResNet [6] introduced residual connections to enable the training of much deeper networks.

In this homework, we will focus on modern CNN architectures known for their effectiveness in face recognition tasks, such as ResNet [6] and its variants [7, 8]. These architectures are designed to extract robust features that are crucial for accurate face recognition.

1.3 Face Recognition Loss Functions

Face recognition presents unique challenges, including variations in lighting, pose, expression, and occlusion, all of which can significantly affect the performance of recognition systems. To address these challenges, the loss functions used to train face recognition models must ensure that the features extracted are not only discriminative but also invariant to these variations.

We will explore several loss functions specifically designed to enhance the performance of face recognition systems, including feature-based losses like Triplet Loss [9] and N-Pair Loss [10], as well as margin-based Softmax losses like ArcFace [2]. These loss functions aim to improve the separability of different identities in the feature space, ensuring that the system performs well even under challenging conditions. To learn more about different types of loss functions please refer to the recitation [Video 1](#) and [Video 2](#) on losses recorded by your TAs.

The network(s) trained on classification will be tested for face verification, where given two face images, your network needs to determine whether these two faces belong to the same person.

2 Problem Specifics

This assignment explores extracting important features from face images that can be used to effectively verify identities, without explicitly training for it. For this, we will focus on following broad parts:

- **Feature extractor:** A face classifier that extracts feature vectors from face images. It consists of two parts:
 - **Objective:** Your goal is to design a model that can learn and identify distinctive facial features (like skin tone, hair color, nose size, etc.) from a person's face image. These features will be represented as a fixed-length feature vector, commonly known as a face embedding.
 - **How It Works:** To achieve this, you will explore different architectures that involve multiple convolutional layers. These layers will help your model break down the image into various components:
 - * **Low-Level Features:** The first few layers might detect simple elements like edges or lines.
 - * **High-Level Features:** As you stack more convolutional layers, the model will start to recognize more complex patterns by combining these low-level features, such as detecting shapes, textures, and specific facial attributes.
 - **Why It Matters:** This hierarchical decomposition of the image is essential because it allows the model to understand and capture the detailed characteristics that make each face unique.
- **Classification Layer:**
 - **Objective:** Once you have obtained the feature vector (face embedding) from the feature extractor, you'll use it to classify the image into one of several categories (e.g., identifying the person in the image from a set of known identities).
 - **How It Works:** The feature vector will be passed through a linear layer or a Multi-Layer Perceptron (MLP), which will output the probability of the image belonging to each of the 'N' categories. The model will then use cross-entropy loss during training to optimize its performance.
 - **Why It Matters:** After training the model, these feature vectors can be used not just for classification but also for face verification tasks, where you'll compare feature vectors from different images to determine if they belong to the same person.

Your model needs to be able to learn facial features (e.g., skin tone, hair color, nose size, etc.) from an image of a person's face and represent them as a fixed-length feature vector called *face embedding*. In order to do this, you will explore architectures consisting of multiple convolutional layers.
- **The verification:** A system (network) that computes the similarity between feature vectors of two images. Essentially, the face verification system takes two images as input and outputs a similarity score that represents how similar the two images are - that is, whether it is the same person or not. The verification consists of the following steps:
 1. Extracting the feature vectors from two (or more) images.
 2. Compare the feature vectors using a similarity metric (example cosine similarity).
 3. [If required] Retrain or finetune the classification model for the verification task.

For a pictorial representation of the problem and its workflow, please refer to ??

3 Data Description

For this homework, your training task (classification) and inference task (verification) would be different. Thus we will have two datasets.

- The first dataset for **classification** training used in this homework is a subset of the [VGGFace2 dataset](#). This dataset is very widely known and used in research and industry. Images are downloaded from Google Image Search and have large variations in pose, age, illumination, ethnicity, and profession (e.g., actors, athletes, politicians). The classification dataset consists of 8631 identities with image resolution of 112×112 .
- The second dataset is a **verification** dataset used only for validation and testing. It consists of 6000 image pairs with 5749 identities in total.

3.1 File Structures

3.1.1 Classification Dataset

- Each sub-folder in **train** and **dev** contains images of one person, and the name of that sub-folder represents their ID.
 - **train**: You are supposed to use the **train** set to train your model **both for the classification task and verification task**.
 - **dev**: You are supposed to use **dev** to validate the classification accuracy.

3.1.2 Verification Dataset

- **ver_data**: This is the folder of all face images with unknown identities.
- **test_pair.txt**: This is the test file where each two sampled images are treated as a pair. You need to predict a score for each pair for the **final_kaggle_score**.

3.2 Dataset Class and Data Loader

The Data set class in this homework is designed to handle classification and verification, the two objectives of this homework. The **ImageDataset** class is designed to handle single images and their corresponding labels. In the initialization, we perform transformations (like resizing and normalization) to augment the data. Additionally, we load image paths and labels from **labels.txt** and ensure a one to one mapping between images and the corresponding labels. The `__getitem__` function returns a single transformed image and its corresponding labels.

The **ImagePairDataset** class is designed for tasks involving pairs of images, such as verification tasks. In the initialization we get a text file containing pairs of image paths and a match label (1 if they match, 0 if they don't). Image pairs are read from the **pairs_file** and we store their match labels. When loading the data loaders, the only thing to keep in mind is to include the image transforms when passing the dataset to ensure you set the right batch size, and that data transformation is applied.

4 CNN Architectures and Data Augmentations

4.1 Introduction

Before diving into the implementation, let's pause and ask an important question:

How do we differentiate faces?

This is a tough question to answer if you are seeing this for the first time, particularly because of the Polyani's paradox about tacit knowledge – “*We know more than we can tell*”. You might think of **facial features** like skin tone, eye shapes, nose size, and other characteristics. These features vary significantly from person to person and are what makes each of us unique. But what features do you pay more/less attention to when you are identifying someone in real life? That is rather challenging to answer.

The primary goal of this assignment is to train a Convolutional Neural Network (CNN) to identify and learn from such important facial features from an image of a person's face, through its training, without us having to explicitly define it. The model will do this by extracting these features and encoding them into a fixed-length vector called a **face embedding**.

Once your CNN model is capable of encoding sufficient and distinctive facial features into these face embeddings, you can then use these embeddings for further tasks, such as:

- **Face Classification:** Assigning an ID or label to a given face.
- **Face Recognition/Verification:** Identifying or verifying a person based on their face. This is a more challenging task, it might be possible that a face that you are trying to verify never appeared in your training set. Ideally, you want your model to be generalizable enough to be able to verify unseen faces!

4.2 How do we train CNNs to produce multi-class classification?

Now comes our second question: how should we train your CNN to produce high-quality face embeddings? It may sound fancy, but conducting *face classification* is just doing a **multi-class classification**: the input to your system is a face's image, and your model needs to predict the ID of the face.

Suppose the labeled dataset contains a total of M images that belong to N different people (where $M > N$). Your goal is to train your model on this dataset to produce “good” face embeddings. You can do this by optimizing these embeddings to predict the face IDs from the images. The resulting embeddings will encode a lot of discriminative facial features, just as desired. This suggests an N -class classification task. A typical multi-class classifier conforms to the following architecture:

Classic multi-class classifier = feature extractor(CNN) + classifier(FC)

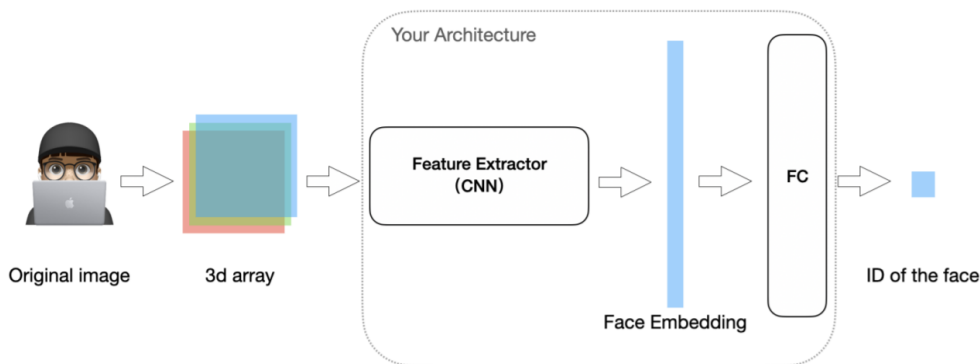


Figure 1: A typical face classification architecture

More concretely, your network consists of several (convolutional) layers for feature extraction. The core operation in a convolutional layer involves sliding a filter (also known as kernel) over the input data (for example an image) to produce a output feature map. The output of the last such feature extraction layers (i.e. the final output feature map) would be the face embedding. You will pass this face embedding through a linear layer whose dimension is *embedding dim* $\times$ *num of face-ids* to classify the image among the N (i.e., num of face-ids) people. You can then use cross-entropy loss to optimize your network to predict the correct person for every training image.

The ground truth will be provided in the training data (making it supervised learning). You are also given a validation set for fine-tuning your model. Please refer to the **Dataset section** where you can find more details about what dataset you are given and how it is organized. To understand how we (and you) evaluate your system, please refer to the **System Evaluation section**.

4.3 Transformations and Data Augmentation

When working with images in deep learning, it's essential to prepare and augment your data to improve your model's performance. PyTorch provides a module called **torchvision.transforms** that is specifically designed for this purpose. This module includes a variety of pre-defined transformations that can be easily applied to images or entire datasets. Some of the most commonly used transformations include **resizing, cropping, flipping, rotating, adjusting brightness/contrast, normalizing pixel values, and converting images to tensors**.

4.3.1 Why Transformations Matter:

Image transformations, often referred to as data augmentation techniques, play a crucial role in training Convolutional Neural Networks (CNNs). While they don't directly generate new features, they offer several significant benefits:

- **1. Increasing Data Volume:** By applying transformations to your images, you effectively increase the size of your dataset. This helps the model learn from a wider variety of examples, which can improve both training and generalization.
- **2. Preventing Overfitting:** Overfitting occurs when a model memorizes the training data instead of learning to generalize. By exposing the model to slightly altered versions of the same images, transformations help the model focus on learning robust features rather than specific details of individual images.
- **3. Invariance:** Transformations teach your model to recognize objects regardless of their orientation, position, or other variations. This means that your model becomes more versatile and capable of handling real-world scenarios where such variations are common.
- **4. Better Generalization:** A diverse training set, created through transformations, exposes your model to a wider range of scenarios. This can lead to better performance when the model encounters new, unseen data.

4.3.2 Important Considerations:

While transformations can be highly beneficial, they should be chosen carefully to suit your specific task. Here are some key points to keep in mind:

- **Appropriate Transformations:** Ensure that the transformations you use are relevant to the type of data you're working with. For instance, if you're working with face images, applying vertical flips might not be appropriate, as upside-down faces are uncommon in real-world data.
- **Impact on Training Time:** Some transformations, especially those that require significant processing, can increase the training time per epoch. It's important to balance the benefits of augmentation with the computational cost.

- **Normalization:** Many models require the input data to be normalized. This involves subtracting the **mean** from each pixel and dividing by the **standard deviation**. This process ensures that the data is on a similar scale, which can help with model convergence. For more on how to normalize images in PyTorch, you can refer to this guide ¹. PyTorch’s `torchvision.transforms.Normalize()` can be used for this purpose, but make sure to check whether it takes in images or tensors as input.

4.3.3 Mixup

Mixup [11] is a powerful data augmentation technique that creates new training samples through linear interpolation of existing data points and their corresponding labels. This method encourages the model to learn smoother decision boundaries and reduces overconfidence in predictions. Given two randomly selected training samples, (x_i, y_i) and (x_j, y_j) , a new, virtual training sample $(\hat{x}, \hat{y})$ is generated as follows:

$$\begin{aligned}\hat{x} &= \lambda x_i + (1 - \lambda)x_j \\ \hat{y} &= \lambda y_i + (1 - \lambda)y_j\end{aligned}$$

Here, λ is a mixing coefficient sampled from a symmetric Beta distribution, $\lambda \sim \text{Beta}(\alpha, \alpha)$, where α is a hyperparameter. The Beta distribution allows for a wide range of mixing proportions, from no mixing ($\lambda = 0$ or $\lambda = 1$) to equal mixing ($\lambda = 0.5$).

The theoretical underpinning of Mixup is based on the principle of Vicinal Risk Minimization (VRM). VRM suggests that augmenting the training data with examples from the "vicinity" of the original data points can improve generalization. Mixup extends this concept by defining the vicinity of a data point as a linear combination with other data points. This encourages the model to learn simpler, linear behavior between training examples, which acts as a form of regularization. By training on these "soft" labels, the model becomes less prone to memorizing the training data and exhibits increased robustness against adversarial examples and noisy labels.

4.3.4 CutMix

CutMix [12] is another data augmentation technique that, like Mixup, combines information from two images. However, instead of blending the pixel values, CutMix cuts a rectangular patch from one image and pastes it onto another. The ground truth labels are then mixed proportionally to the area of the patch that was pasted.

The generation of a new sample $(\tilde{x}, \tilde{y})$ from two input samples (x_a, y_a) and (x_b, y_b) is defined as:

$$\begin{aligned}\tilde{x} &= \mathbf{M} \odot x_a + (\mathbf{1} - \mathbf{M}) \odot x_b \\ \tilde{y} &= \lambda y_a + (1 - \lambda)y_b\end{aligned}$$

In this formulation, $\mathbf{M} \in \{0, 1\}^{W \times H}$ is a binary mask indicating where to drop and fill pixels from the two images, and $\mathbf{1}$ is a binary mask filled with ones. The $\odot$ symbol represents element-wise multiplication. The mixing ratio, λ , is determined by the proportion of the area of the patch relative to the total image area. The bounding box coordinates for the patch are sampled uniformly, which in turn determines the mask $\mathbf{M}$ and the value of λ .

CutMix offers the advantage of not removing informative pixels, a drawback of regional dropout methods like Cutout. By forcing the model to recognize two objects from a single image, it improves the model’s localization ability and encourages it to learn from less discriminative parts of an object. This makes CutMix particularly effective for tasks like object detection and weakly supervised localization, often outperforming Mixup and other augmentation methods in these areas.

¹[How to Normalize Images in PyTorch](#)

4.4 Create deeper layers with residual networks

Having a network that is good at feature extraction and being able to efficiently train that network is the core of the classification task. This homework requires training deep neural networks and, as it turns out, deep neural networks are difficult to train, because they suffer from vanishing and exploding gradients types of problems. Here we will learn about **skip connections** that allow us to take the activations of one layer and suddenly feed it to another layer, even much deeper in the network. Using that, we can build **residual networks** (like ResNets), which enable us to train very deep neural networks, sometimes even networks of over one hundred layers.

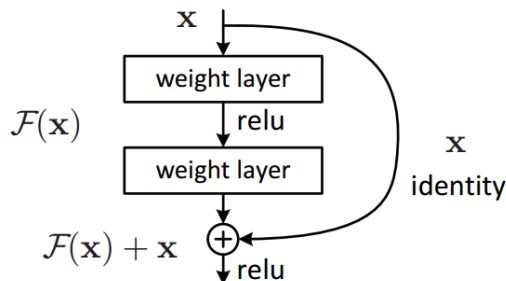


Figure 2: A Residual Block

Resnets are made of something called residual blocks, which are a set of layers that are connected to each other, and the input of the first layer is added to the output of the last layer in the block. This is called a **residual connection**. This identity mapping does not have any parameters and is just there to add the input to the output of the block. This allows deeper networks to be built and trained efficiently. Several other blocks make use of residual blocks and residual connections and can be used for the classification task, such as **ResNet**, **SEResNet**, **ConvNeXt**, **MobilNet**, **etc.** *You are encouraged to read their respective research papers to understand better how they work and implement blocks from these architectures.*

4.4.1 ResNet

ResNet models were proposed in “[Deep Residual Learning for Image Recognition](#)”. Here we have the 5 versions of ResNet models, which contain 18, 34, 50, 101, and 152 layers, respectively. Detailed model architectures can be found in the paper linked above.

4.4.2 ConvNeXt

ConvNeXt is a recent CNN architecture that uses inverted bottlenecks inspired by the Swin Transformer, residual blocks, and depthwise separable convolutions instead of regular convolutions. A comparison of the ResNet-50 and ConvNeXt-T and the detailed architecture can be found in “[A ConvNet for the 2020s](#)”.

Since you will be using blocks and customized versions of these architectures, the performance may or may not match the expected outcomes based on the benchmarks in the papers. Thus, you are encouraged to explore various architectures systematically to get you to the high cut-off. That’s pretty much everything you need to know for your Classification Kaggle competition. Go for it!

5 Face Recognition

To develop a robust face recognition system, it is essential to understand how the training data and model architecture are defined and interconnected.

Let us define the training dataset as $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i \in [N]}$, where $\mathbf{x}_i \in \mathbb{R}^{H \times W \times C}$ indicates an input face image, $\mathbf{y}_i$ is the corresponding one-hot ground-truth identity for the image, i.e., a one-hot class label, and N denotes the dataset size. We can define our face recognition system as the stack of a CNN-based feature extractor $\mathbf{z} = f_{\theta}(\mathbf{x})$ with learnable parameters θ , which transforms the input images $\mathbf{x}$ into features $\mathbf{z} \in \mathbb{R}^D$, where D indicates the dimension of the feature space, and a linear projection head $\mathbf{s} = g_{\omega}(\mathbf{z})$ with learnable parameters ω , which maps the features $\mathbf{z}$ into scores $\mathbf{s} \in \mathbb{R}^C$ with dimension C .

To express the relationship between the feature vectors and the scores in matrix form, we can write:

$$\begin{aligned}\mathbf{z}_i &= f_{\theta}(\mathbf{x}_i) \\ \mathbf{s}_i &= g_{\omega}(\mathbf{z}_i) = \mathbf{z}_i W\end{aligned}$$

Here, $W \in \mathbb{R}^{D \times C}$ is the weight matrix of the linear projection head, which maps the feature vector $\mathbf{z}_i \in \mathbb{R}^D$ to the score vector $\mathbf{s}_i \in \mathbb{R}^C$. Each element s_i^j in the score vector represents the unnormalized logit for class j , which will be used to compute the probability of the input image belonging to each identity class through the Softmax function.

5.1 Training Face Classifiers with Softmax Cross-Entropy

In this part, we will show the first modeling of a face recognition system as a **face identity classifier**. As you learned from the class, this is as easy as solving a classification problem using a feature extractor of CNN and a classification head with learnable weights $W \in \mathbb{R}^{D \times C}$ *(Here we omit the bias term in the linear layer, but it can be easily incorporated into W)*, can be done by optimizing the Cross-Entropy (CE) loss with the Softmax function. Recall that, in the classification setting, we usually term the score vectors $\mathbf{s}$ as **logits**, and the Softmax function converts them into probabilities:

$$\mathbf{p} = \text{Softmax}(\mathbf{s}) = \left[\frac{\exp(s^1)}{\sum_{k=1}^C \exp(s^k)}, \dots, \frac{\exp(s^C)}{\sum_{k=1}^C \exp(s^k)} \right] \in \mathbb{R}^C,$$

where each element in the vector indicates the (predicted) probability of this sample belong to a class, e.g., $p^j = \frac{\exp(s^j)}{\sum_{k=1}^C \exp(s^k)}$ is the (predicted) probability of a sample belonging to class k . We optimize the Cross-Entropy (CE) to encourage the probability of each sample $\mathbf{x}_i$ belonging y_i to be maximized, i.e, the cross-entropy loss between the predicted probability and the one-hot ground truth to be minimized:

$$\mathcal{L}_{\text{CE}} = \frac{1}{N} \sum_i^N \sum_j^C - y_i^j \log p_i^j$$

With proper CNN architecture and data augmentation, we can obtain a strong classification baseline for this homework.

5.2 Limitations of Treating Face Recognition as a Classification Problem

There are two primary limitations when treating face recognition as a standard classification problem:

1. Closed-set Recognition Limitation:

- A classifier trained for face recognition can only identify faces with labels y that were present in the training data.

²,

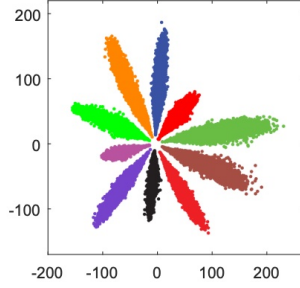


Figure 3: Feature visualization of Softmax CE loss.

During inference, the model predicts the identity of a given test face by selecting the class with the highest predicted probability:

$$\hat{y} = \arg \max_j (p^j)$$

where p^j represents the predicted probability for class j (i.e., identity j). This means the system operates under a **closed-set assumption**, where it can only recognize faces from a fixed set of known identities.

If we introduce a new identity not present in the training set (an **open-set identity**), the model cannot recognize it unless the entire system is retrained with this new data. This retraining process is computationally expensive and impractical for large-scale face recognition systems involving millions or even billions of identities, as is common in real-world applications. One can solve this by bypassing the classification layer and relying solely on the **feature extractor**. Instead of predicting class probabilities, the model generates a feature representation (embedding) of the input face. Recognition is then performed by comparing this feature with those of the training faces:

$$\hat{y} = \arg \max_{y_i} (\text{similarity}(\mathbf{z}, \mathbf{z}_i))$$

Here:

- $\mathbf{z}$ represents the feature vector of the test face.
- $\mathbf{z}_i$ is the feature vector of the i -th training sample.
- $\text{similarity}(\mathbf{z}, \mathbf{z}_i)$ is a similarity measure (e.g., cosine similarity) between the test and training features.

The model assigns the identity of the training sample that has the **highest similarity** to the test sample. This approach enables **open-set recognition**, allowing the system to generalize to new identities without retraining.

2. Poor Feature Representation for Similarity:

- When the feature extractor is trained using standard classification techniques, specifically, Cross-Entropy (CE) loss combined with a Softmax activation, the resulting features are often not optimized for direct similarity comparisons.

In classification tasks, the model focuses on maximizing the correct class probability rather than ensuring that features of the same identity are close in the feature space and those of different identities are far apart. As a result, the extracted features may not provide a reliable similarity measure for face recognition tasks. We will delve deeper into this limitation and its implications in section 5.3, where we analyze how CE loss affects the discriminative power of the extracted features.

Thinking Note

Prompt Engineering with AI – Shift the task definition:

- “Why does high classification accuracy imply good face verification performance?”
- “Why does high classification accuracy not imply good face verification performance?”

The second answer stops treating accuracy as the goal.

5.3 A Closer Look: What Are We Learning with CE?

$$\begin{aligned}
 l_{\text{CE}} &= \sum_j^C -y^j \log p^j \\
 &= \sum_j^C -y^j \log \frac{\exp(s^j)}{\sum_{k=1}^C \exp(s^k)} \\
 &= \log \left(1 + \sum_{k=1, k \neq y}^C \exp(s^k - s^y) \right) \\
 &= \log \left(1 + \exp \left(\log \left(\sum_{k=1, k \neq y}^C \exp(s^k - s^y) \right) \right) \right) \\
 &\geq \max \left(\log \left(\sum_{k=1, k \neq y}^C \exp(s^k - s^y) \right), 0 \right) \\
 &\geq \log \sum_{k=1, k \neq y}^C \exp(s^k) - s^y \\
 &\approx \max_{k \in [C], k \neq y} (s^k) - s^y
 \end{aligned}$$

The objective of Softmax CE loss: optimizing the ground-truth score s^y to be larger than the maximum of non-ground-truth score $\max_{k \in [C], k \neq y} (s^k)$.

We have $s^j = \mathbf{z} \mathbf{w}_j^T = \|\mathbf{z}\| \|\mathbf{w}_j^T\| \cos \theta_j$. This indicates that Softmax CE loss will strengthen the length of feature vector and the weight vector, resulting in a radial feature space, as shown in fig. 3. A radial feature space presents very limited discriminative features for feature matching of face recognition. For example, $\mathbf{z}_1$ and $\mathbf{z}_2$ belong to the same class, but it present small feature similarity compared to $\mathbf{z}_1$ and $\mathbf{z}_3$ due to the strengthened features.

Thinking Note

Prompt Engineering with AI – Change what is being optimized:

- “What does Softmax Cross-Entropy optimize?”
- “What does Softmax Cross-Entropy optimize with respect to the feature space?”

The explanation changes when attention shifts beyond the logits.

5.4 Objectives of Face Recognition

The primary goals for an effective open-set face recognition system are as follows:

- **Maximize Intra-Class Similarity:** Ensure that face images of the same identity have highly similar feature representations. This means that features extracted from different images of the same person should be close together in the feature space, which facilitates accurate identification.

- **Minimize Inter-Class Similarity:** Ensure that face images of different identities have distinct feature representations. This involves making the features of different individuals sufficiently different, thereby reducing the chance of mis-identification.
- **Maintain a Large Margin between Intra-Class and Inter-Class Similarity:** Ensure that the similarity between features of the same class (intra-class similarity) is significantly higher than the similarity between features of different classes (inter-class similarity). This is crucial for robust face recognition, especially when dealing with challenging cases such as similar-looking images taken under varying conditions.

To mathematically capture these objectives, we define a loss function that penalizes small intra-class distances and large inter-class distances. A common formulation is:

$$\mathcal{L} = \max(s^n - s^p + m, 0),$$

where:

- $s^p = \text{Similarity}(\mathbf{z}_i, \mathbf{z}_j^p)$ represents the similarity score between the anchor feature $\mathbf{z}_i$ and a positive feature $\mathbf{z}_j^p$ from the same identity.
- $s^n = \text{Similarity}(\mathbf{z}_i, \mathbf{z}_k^n)$ represents the similarity score between the anchor feature $\mathbf{z}_i$ and a negative feature $\mathbf{z}_k^n$ from a different identity.
- m is a margin parameter that enforces a gap between the positive and negative similarity scores.

The objective of this loss function is to ensure that s^n (the similarity with a negative sample) is at least m units less than s^p (the similarity with a positive sample). If this condition is not met, the loss will be positive and the model will be penalized, thereby pushing the features to satisfy this margin constraint. By optimizing this loss, the model learns to create a feature space where same-identity faces are clustered together, and different-identity faces are well separated. Also, from this objective, we can observe that in purely CE training, the loss functions only encourage the positive similarity s^y to be larger than the maximal of the negative similarities $s^k, k \in [C], k \neq y$, without explicitly encourage a margin.

In the next, we will revisit the two paradigms of achieving this objective: feature-based loss functions (also known as metric learning) and margin-based loss functions. We will also go through a unified view of both paradigms at the end.

5.5 Feature-based Loss Functions

5.5.1 Overview

Feature-based loss functions, also known as metric learning losses, focus on directly optimizing the feature space to enhance the discriminative power of the learned representations. Instead of merely classifying faces into predefined categories, these loss functions aim to structure the feature space such that similar identities are clustered closely together while dissimilar identities are pushed far apart. This approach is particularly advantageous for open-set face recognition, where the system encounters identities not seen during training.

5.5.2 Triplet Loss

Triplet loss [9] is one of the most widely used feature-based loss functions. It operates on triplets of images: an anchor image $\mathbf{x}_i$, a positive image $\mathbf{x}_i^p$ (from the same identity as the anchor), and a negative image $\mathbf{x}_i^n$ (from a different identity). The goal of the triplet loss is to ensure that the distance between the anchor and the positive image is smaller than the distance between the anchor and the negative image by at least a margin α . The triplet loss function is defined as:

$$\mathcal{L}_{\text{triplet}} = \sum_{i=1}^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + m \right]_+,$$

where $f(x)$ denotes the feature representation of image x learned by the model, $\|\cdot\|_2^2$ is the squared Euclidean distance (Cosine similarity can also be used, in which case the sign of the similarity score would be flipped), and $[\cdot]_+$ denotes the hinge loss that only penalizes the loss when it is positive.

This loss encourages the model to learn a feature space where:

- The features of images from the same identity (anchor and positive) are close together.
- The features of images from different identities (anchor and negative) are far apart by at least the margin m .

Looking closer at the Triplet loss function, you will find it is very similar to what was defined earlier as the objective for face recognition, i.e., section 5.4, by replacing the similarity metric as the Euclidean distance and flipping the sign.

5.5.3 N-Pair Loss

N-Pair loss [10] generalizes the concept of triplet loss by considering multiple negative samples for each anchor-positive pair, rather than just one. This allows for more robust training by incorporating a broader context of the feature space. The N-Pair loss is defined as:

$$\mathcal{L}_{\text{N-Pair}} = \sum_{i=1}^N \log \left[1 + \sum_{j \neq i} \exp \left(f(x_i^a)^T f(x_j^n) - f(x_i^a)^T f(x_i^p) \right) \right],$$

where $f(x_i^a)^T f(x_j^n)$ denotes the similarity between the anchor and a negative sample, and $f(x_i^a)^T f(x_i^p)$ denotes the similarity between the anchor and the positive sample.

The N-Pair loss effectively encourages the model to maximize the similarity between the anchor and the positive pair while minimizing the similarity with **multiple negatives** (compared to only one negative in triplet loss), providing a more comprehensive training signal.

5.5.4 Remark

Feature-based loss functions are powerful for structuring the feature space, but they come with certain challenges:

- **Directly Optimizing the Objective:** You will find these metric loss directly optimize the objectives we listed earlier.
- **Difficult to Tune:** Choosing the right margin parameter α in triplet loss or selecting the appropriate number of negative samples in N-Pair loss can be challenging and may require extensive hyper-parameter tuning. Usually, **it makes life easier to combine the metric learning loss with CE loss**.
- **Hard Sample Mining:** The effectiveness of feature-based losses heavily relies on the selection of hard samples—triplets or pairs where the model struggles to differentiate between positive and negative samples. Efficient mining strategies are crucial but can be computationally expensive.
- **Training Complexity:** These loss functions often require careful batching and sampling strategies to ensure the training process is stable and effective, adding complexity to the model training pipeline.

Despite these challenges, feature-based loss functions are indispensable for applications requiring robust face recognition, especially in open-set scenarios where the system must generalize to unseen identities.

5.6 Margin-based Softmax Loss Functions

5.6.1 Overview

Margin-based Softmax loss functions are designed to enhance the discriminative power of features learned by deep neural networks, particularly in classification tasks like face recognition. These loss functions modify the traditional Softmax Cross-Entropy (CE) loss by introducing a margin that separates different classes more distinctly in the feature space. This approach is particularly useful in scenarios where fine-grained distinctions between classes (e.g., different identities) are critical, as it encourages the model to maximize inter-class variance while minimizing intra-class variance.

5.6.2 ArcFace

ArcFace is one of the most popular margin-based Softmax loss functions. It introduces an angular margin to the Softmax function, which helps in learning more discriminative features by pushing the decision boundaries further away from each other. The formulation of ArcFace is as follows:

Given the feature vector $\mathbf{z}_i$ of a sample and the corresponding weight vector $\mathbf{w}_j$ for class j , the original Softmax function is defined as:

$$s^j = \mathbf{z}_i \cdot \mathbf{w}_j = \|\mathbf{z}_i\| \|\mathbf{w}_j\| \cos \theta_j,$$

where θ_j is the angle between the feature vector $\mathbf{z}_i$ and the weight vector $\mathbf{w}_j$.

ArcFace modifies this by adding an **angular margin** m to θ_j , effectively making it harder for the model to classify samples correctly, thus pushing the model to learn more discriminative features. To make the angular margin effective, the magnitude of both feature and weight vectors are normalized to unit length and scaled by a fixed parameter α , which transforms the logit (similarity score) to:

$$s^j = \gamma \cos(\theta_j + m) = \alpha (\cos \theta_j \cos m - \sin \theta_j \sin m).$$

The modified logits are then passed through the Softmax function as usual:

$$\mathbf{p} = \text{Softmax}(s^j),$$

The key idea is that by introducing the margin m , the model is forced to learn features that not only separate classes, but also create a margin between them in the angular space, leading to more robust face recognition.

5.6.3 Combined Margin-based Loss Functions

In addition to ArcFace, other margin-based loss functions such as CosFace [1], SphereFace [13], and AM-Softmax [14] have been proposed, each introducing different types of margin (additive angular, multiplicative angular, etc.). These can be combined to create a unified margin-based loss function that takes advantage of multiple margin types simultaneously.

For instance, a combined margin-based loss function could be formulated as:

$$s^j = \gamma (\cos(\theta_j + m_1) + m_2),$$

where m_1 and m_2 are different types of margins (e.g., angular and additive). This allows for more flexibility in training and can lead to improved performance in specific face recognition tasks. You are encouraged to try different combination of margin and find out which one works the best.

5.6.4 Remark

Margin-based Softmax loss functions have several advantages:

- **Easier to Train:** Compared to feature-based loss functions, margin-based Softmax losses are easier to implement and train because they do not require complex sample mining strategies. They directly modify the logits before applying the Softmax function, which simplifies the training pipeline.
- **Effective for Fine-Grained Classification:** By introducing a margin, these loss functions make the decision boundaries between classes more distinct, which is crucial for tasks like face recognition where classes (identities) can be very similar.
- **Limited to One Positive Class:** A limitation of these methods is that they are designed for single-label classification scenarios, where each sample belongs to only one class. They are less effective in multi-label scenarios or when the task requires recognizing multiple faces simultaneously. In addition, recall that methods such as cutmix and mixup do some kind of label mixing, which may not work well with margin based loss functions.
- **Scalability:** Margin-based Softmax loss functions scale well with the number of classes, making them suitable for large-scale face recognition systems.

In summary, margin-based Softmax loss functions strike a balance between ease of training and the ability to learn highly discriminative features, making them a popular choice for modern face recognition systems.

5.7 A Unified View of Feature-based and Margin-based Softmax Loss Functions

5.7.1 Positives vs. Negatives

Both feature-based and margin-based Softmax loss functions share a common goal: to enhance the discriminative power of the learned features by maximizing the similarity between samples of the same class (positives) and minimizing the similarity between samples of different classes (negatives). While they approach this objective differently, the underlying principles are closely related.

Feature-based loss functions, such as Triplet Loss and N-Pair Loss, explicitly operate on the distances or similarities between anchor-positive and anchor-negative pairs. They directly optimize the feature space to ensure that positive pairs are close and negative pairs are far apart.

Margin-based Softmax loss functions, like ArcFace, introduce a margin in the classification layer to implicitly achieve a similar effect. By modifying the logits before applying the Softmax function, these losses create a separation (or margin) between different classes, pushing the model to learn more distinct features. The weight of the last linear classifier can be viewed as an anchor for each classes (especially after being normalized), which thus ties the margin-based Softmax loss functions closely with feature based loss functions. In the next, we will introduce more modern loss functions that provides a unified view of both paradigms.

5.7.2 Circle Loss

Circle Loss [15] is an example of a unified approach that combines elements of both feature-based and margin-based losses. It introduces a unified perspective by directly optimizing the similarity scores with a margin, while also ensuring that the optimization is focused on hard samples (those that are difficult to classify). Circle Loss is defined as:

$$\mathcal{L}_{\text{circle}} = \log \left[1 + \sum_{\text{pos}} \exp(\gamma(\cos \theta_{\text{pos}} - m)) \sum_{\text{neg}} \exp(\gamma(\cos \theta_{\text{neg}} + m)) \right],$$

where γ is a scaling factor, m is the margin, θ_{pos} and θ_{neg} are the angles corresponding to positive and negative pairs, respectively.

Circle Loss unifies the optimization of intra-class and inter-class similarities by applying different margins to positive and negative pairs, ensuring that the learned feature space is well-structured for discriminative tasks like face recognition.

5.7.3 Loss Function Rewrite

Both feature-based and margin-based losses can be rewritten to highlight their commonalities. For instance, the triplet loss can be seen as a special case of a margin-based loss when we view the distance metric as a form of similarity score with a margin. Similarly, margin-based Softmax losses can be interpreted as imposing a margin directly on the logit-level comparisons that occur in feature-based losses.

This unified perspective allows us to see that both approaches are variations of the same fundamental principle: enforcing a margin between positive and negative samples to improve feature discriminability.

5.7.4 Supervised Contrastive Loss

Supervised Contrastive Loss (SupCon) [16] is another loss function that unifies the principles of feature-based and margin-based approaches. It leverages contrastive learning, a self-supervised technique, in a supervised setting to enhance intra-class compactness and inter-class separability. The SupCon loss is defined as:

$$\mathcal{L}_{\text{supcon}} = \frac{1}{N} \sum_{i=1}^N \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_p / \tau)}{\sum_{a=1}^{2N} \mathbb{I}_{[a \neq i]} \exp(\mathbf{z}_i \cdot \mathbf{z}_a / \tau)},$$

where $\mathbf{z}_i$ and $\mathbf{z}_p$ are the feature representations of the anchor and positive samples, τ is a temperature scaling parameter, and $P(i)$ denotes the set of positives for anchor i .

Supervised Contrastive Loss encourages samples from the same class to be pulled together in the feature space, while samples from different classes are pushed apart. This aligns with the goals of both feature-based and margin-based losses, providing a robust approach to learning discriminative features.

5.8 Remark

Although these loss functions provide a unified view of both learning paradigms, they usually require a very large batch size (>2048) to achieve reasonable performance. We suggest that you try these loss functions with distributed training over multiple GPU or in combination with previous loss functions.

You should be able to achieve a high enough score with previous loss functions as well.

6 Face Verification

Now let us switch gears to face verification. After training of your model using the loss functions, the input to your system will now be a pair of face images that may or may not belong to the same person during inference. Given a *pair*, your goal is to output a numeric score that quantifies how similar the faces in the two images are. A higher score indicates a higher confidence about whether the faces in the two images are of the same person.

6.1 Building upon the multi-class classification

If your model yields high accuracy in face classification, you **might** already have a good Feature Extractor for free. That being said, if you remove the fully connected/linear layer, this leaves you with a CNN that “can” (*probably could* would be more accurate here) generate discriminative face embeddings, given arbitrary face images.

6.2 Verification Pipeline

We shall all agree that The face embeddings of the same person should be similar (i.e., the distance between feature vectors generated is small), even if they are extracted from different images. After being trained explicitly to maximize the similarity of positive pairs, minimize the similarity of negative pairs, and encourage some margin between positives and negatives, your model is capable to generate accurate face embeddings. We only need to compute a proper **distance metric** to evaluate how close the given face embeddings are. If two face embeddings are close in distance, they are more likely to be from the same person.

Thinking Note

Prompt Engineering with AI – Swap the comparison operator:

- “Why do we use cosine similarity for face verification?”
- “Why do we use cosine similarity instead of Euclidean distance for face verification?”

The comparison changes once distance is made explicit.

If you follow this design, your system should look like the Figure below. Please notice that the Feature Extractor in Figure 4 is the same one, even though it is drawn twice.

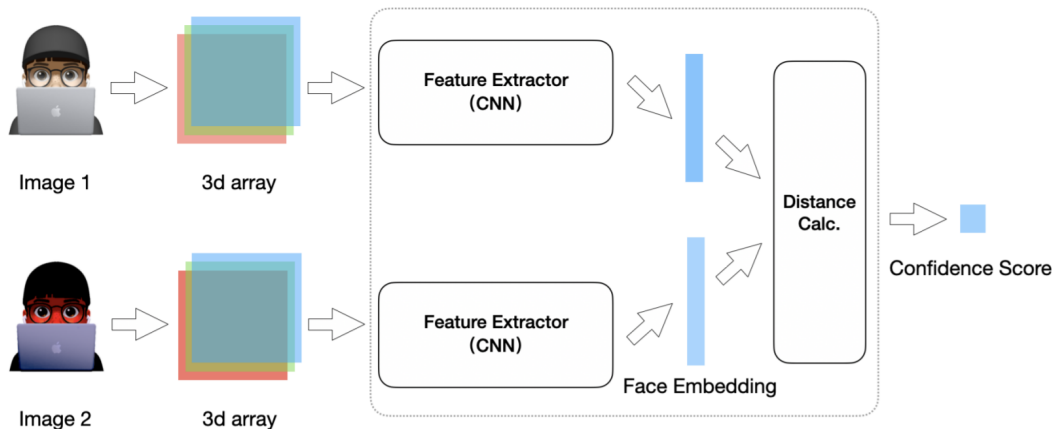


Figure 4: Face verification architecture

7 Kaggle Competition Details

For this assignment, you will be submitting a unified csv file on a **Kaggle** competition for the both the face classification and verification tasks. The purpose of this dual-mode competition is for you to understand how CNNs enable learning for the two tasks that seem similar, but fundamentally differ, and how transfer learning connects the two. It is important to get a sense of the capabilities and limitations of transfer learning - how higher classification tasks help in verification but at the same time higher classification accuracy may not necessarily imply a good verification system (and vice-versa, as shown in CE loss).

- **Task 1: Face classification**

Goal: Given a person's face, return the identity of the face. This is a closed set training/evaluation task. Identities are seen during training.

- **Task 2: Face verification**

Goal: Given a list of known and unknown identities, map each identity to a known or unknown (or "no-correspondence") label. This is an open set training/evaluation task. Identities are NOT seen during training.

- **Note:** You can make multiple submissions to the Kaggle competition, one for each training. However, you can only submit one unified csv file at a time by merging two csv files, one for each task. The code to merge is provided in the starter notebook. Details on the evaluation metrics can be found in [section 8](#).

8 Evaluation System

This assignment will be scored out of 100 pts, but with a small catch to encourage early checkpoint submission. If you make a successful checkpoint submission (see instructions in [section 9](#)), you can get all of the 100 points, if you cross the **high cutoff**. If you miss the checkpoint submission deadline, we will lower your score by **3%** of your final score; TLDR:

- Final score (**without** checkpoint submission): $\frac{\text{your_final_kaggle_score}}{\text{kaggle_leaderboard_score}} \times 97$
- Final score (**with** checkpoint submission): $\frac{\text{your_final_kaggle_score}}{\text{kaggle_leaderboard_score}} \times 100$

The final score will be calculated as a linear combination of the classification and verification scores and the best of the On-Time and/or Slack Submission will be considered.

8.1 Metrics

- **Face Classification**

This is straightforward,

$$\text{Accuracy} = \frac{\# \text{ correctly classified images}}{\text{total images}}$$

- **Face Verification**

In this task, the performance is evaluated using the Equal Error Rate (EER). The EER is the point at which the rate of false acceptances (False Acceptance Rate, FAR) equals the rate of false rejections (False Rejection Rate, FRR). These rates are defined as follows:

$$\text{FAR} = \frac{\# \text{ of false acceptances}}{\text{total } \# \text{ of impostor attempts}}$$

$$\text{FRR} = \frac{\# \text{ of false rejections}}{\text{total } \# \text{ of genuine attempts}}$$

The EER is the value where these two rates are equal, indicating the threshold at which the system's error rates are balanced. A lower EER indicates better performance.

$$\text{EER} = \text{FAR} = \text{FRR}$$

- **Final Kaggle Score**

This will be calculated based on a linear combination of classification and verification scores, using the following equation:

$$\text{Final Kaggle Score} = \alpha \times \text{Accuracy} + \beta \times (1 - \text{EER})$$

where $\alpha = 0.5$ and $\beta = 0.5$

8.2 On-Time Submission:

Performance cut-offs for the Kaggle competition will be released on Piazza soon after the checkpoint deadline has passed. We will release the following cutoffs with the corresponding point values:

- High (100/100)
- Medium (80/100)
- Low (60/100)
- Very Low (40/100)

Scores will be interpolated linearly between these cutoffs.

8.3 Slack Submissions:

- **Scores submitted to the original Kaggle Competition after the final deadline will NOT be considered, no matter how many slack days a student has left.** A student wishing to submit a P2 score after the final deadline must join and continue by submitting to the SLACK Kaggle Competition instead (link is provided on the first page of this writeup).
- A student is granted up to 10 Slack days, which they may use across all P2 submissions (and no other type of assignments) at their discretion throughout the semester. Once a student runs out of Slack days, the best viable submission before the student ran out of slack days will be considered.
- Students are in charge of tracking their own usage of Slack days.
- As mentioned, submissions made to the slack competition will be ineligible for the bonus points even though high cutoff was achieved by the original final deadline.

8.4 Bonus Points:

Bonus points will be assigned ONLY if a student successfully makes an early checkpoint submission AND the on-time submission exceeds the high cutoff. Bonus points are calculated by linearly interpolating between the high cutoff and the highest individual's private score. The highest performance will get the full 3 bonus points. Bonus points are calculated and assigned at the end of the semester.

9 Checkpoint Submissions Instructions

The checkpoint submission is intended to encourage early submissions and it will consist of two tasks that need to be completed on or before the checkpoint submission deadline. The Checkpoint Submission on Gradescope and Canvas Quiz may be completed asynchronously in any order:

- Complete the **P2 Starter Notebook** and make your first submission to the Kaggle competition to pass the low cutoff on or before the checkpoint submission deadline.
- Complete the **Homework 2 Quiz** on Canvas.

10 Final (On-Time/Slack) Submission Instructions

This will be a Gradescope submission consisting of the following (please follow the steps in the "Submission Instructions" part of the student notebook):

- Step 1: Generate Your Model Metadata File
- Step 2: Complete the README Section
- Step 3: Provide Required Credentials
- Step 4: Set File Paths Correctly
- Step 5: (Optional) Add any additional files
- Step 6: Generate and submit the final zip file to Gradescope

Contact your TA in case you struggle at this step.

11 Conclusion

This homework will help you develop fundamental understanding of one of the foundational models of deep learning, the convolutional neural network, for applications in face verification. It is important that you understand the dual training objective nature (classification and verification) of this homework along with how transfer learning plays a key role in connectin these objectives. We strongly recommend attending [bootcamp for HW2P2](#) and Office Hours make the most of this.

This is the end of HW2P2, and the beginning of a new convoluted view of the world.

Good luck and have fun, deep learner!

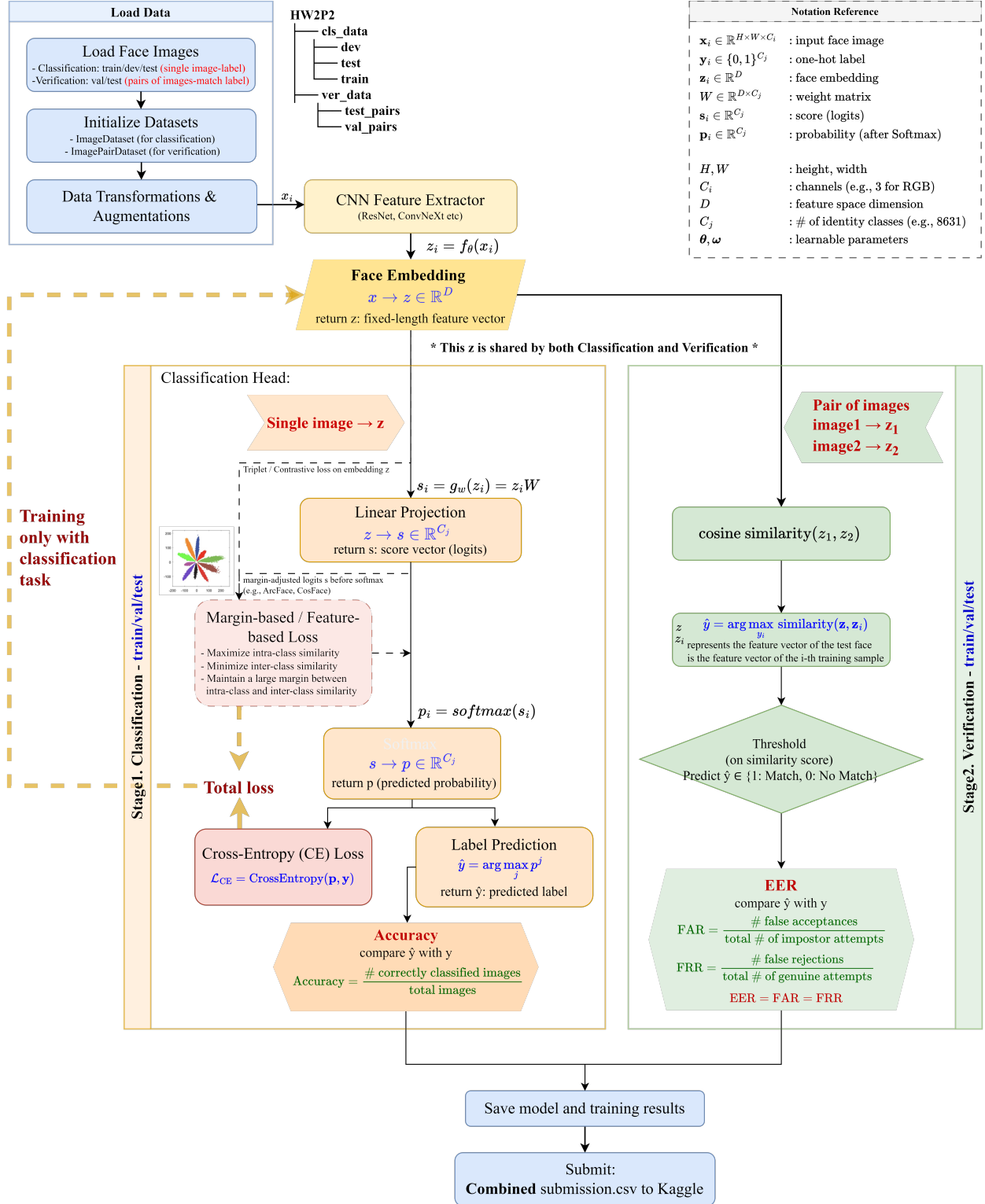


Figure 5: Flowchart for HW2P2

12 Appendix

12.1 List of relevant recitations

Please review the below recitations for supplementary material that could be helpful for this assignment -

- Pytorch Fundamentals
- OOPS Fundamentals
- Google Colab
- GCP
- Kaggle
- Data Loaders
- WandB
- Blocks coding
- Discriminative Losses

References

- [1] Hao Wang, Yitong Wang, Zheng Zhou, Xing Ji, Dihong Gong, Jingchao Zhou, Zhifeng Li, and Wei Liu. Cosface: Large margin cosine loss for deep face recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5265–5274, 2018.
- [2] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4690–4699, 2019.
- [3] Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11):2278–2324, 1998.
- [4] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. *Advances in neural information processing systems*, 25, 2012.
- [5] Karen Simonyan and Andrew Zisserman. Very deep convolutional networks for large-scale image recognition. *arXiv preprint arXiv:1409.1556*, 2014.
- [6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.
- [7] Jie Hu, Li Shen, and Gang Sun. Squeeze-and-excitation networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 7132–7141, 2018.
- [8] Zhuang Liu, Hanzi Mao, Chao-Yuan Wu, Christoph Feichtenhofer, Trevor Darrell, and Saining Xie. A convnet for the 2020s. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 11976–11986, 2022.
- [9] Elad Hoffer and Nir Ailon. Deep metric learning using triplet network. In *Similarity-based pattern recognition: third international workshop, SIMBAD 2015, Copenhagen, Denmark, October 12-14, 2015. Proceedings 3*, pages 84–92. Springer, 2015.
- [10] Kihyuk Sohn. Improved deep metric learning with multi-class n-pair loss objective. *Advances in neural information processing systems*, 29, 2016.
- [11] Hongyi Zhang, Moustapha Cisse, Yann N. Dauphin, and David Lopez-Paz. mixup: Beyond empirical risk minimization, 2018.

- [12] Sangdoo Yun, Dongyoon Han, Seong Joon Oh, Sanghyuk Chun, Junsuk Choe, and Youngjoon Yoo. Cutmix: Regularization strategy to train strong classifiers with localizable features, 2019.
- [13] Weiyang Liu, Yandong Wen, Zhiding Yu, Ming Li, Bhiksha Raj, and Le Song. Sphreface: Deep hypersphere embedding for face recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 212–220, 2017.
- [14] Feng Wang, Jian Cheng, Weiyang Liu, and Haijun Liu. Additive margin softmax for face verification. *IEEE Signal Processing Letters*, 25(7):926–930, 2018.
- [15] Yifan Sun, Changmao Cheng, Yuhao Zhang, Chi Zhang, Liang Zheng, Zhongdao Wang, and Yichen Wei. Circle loss: A unified perspective of pair similarity optimization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 6398–6407, 2020.
- [16] Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. *Advances in neural information processing systems*, 33:18661–18673, 2020.